

Pauli Matrices

Fundamental operators for two-level quantum systems

Two-level quantum systems require operators acting on a two-dimensional Hilbert space. The most general Hermitian operator on $\mathbb{C}^2$ contains four real parameters.

The identity matrix and the three Pauli matrices form a complete, orthogonal basis for this space. Because of this, Pauli matrices appear naturally in the description of qubits, spin- $\frac{1}{2}$ particles, and any two-state quantum system.

Why Pauli matrices appear in two-level systems

A general Hermitian 2×2 operator has the form:

$$H = \begin{pmatrix} a & c \\ c^* & b \end{pmatrix}, \quad a, b \in \mathbb{R}, \quad c \in \mathbb{C}$$

This operator contains four real degrees of freedom. The following four matrices form a complete basis:

$$I, \quad \sigma_x, \quad \sigma_y, \quad \sigma_z.$$

This decomposition will become essential when describing **qubit Hamiltonians**, **Bloch sphere geometry**, and **single-qubit gates**.

Definition of the Pauli matrices

$$\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

These matrices are: * Hermitian * Unitary * Traceless * Determinant=-1 * Involutive: $\sigma_i^2 = I$

They represent the three fundamental directions in the state space of a qubit.

Measurement bases associated with Pauli matrices

Each Pauli matrix corresponds to a measurement in a specific orthonormal basis:
* σ_z : computational basis

$$|0\rangle, \quad |1\rangle.$$

* σ_x : symmetric/anti-symmetric superpositions

$$|+\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}}.$$

* σ_y : superpositions with a relative phase

$$|\pm\rangle = \frac{|0\rangle \pm i|1\rangle}{\sqrt{2}}.$$

These three axes will become the x , y and z directions of the Bloch sphere.

Commutation and anti-commutation: a brief conceptual window

Commutator

$$[A, B] = AB - BA$$

The commutator measures whether two operations influence each other. If $[A, B] \neq 0$, **the order of operations matters**. For Pauli matrices:

$$[\sigma_i, \sigma_j] = 2i\epsilon_{ijk}\sigma_k.$$

This structure is identical to the algebra of angular momentum in quantum mechanics.

Anti-commutator

$$\{A, B\} = AB + BA$$

Pauli matrices satisfy:

$$\{\sigma_i, \sigma_j\} = 2\delta_{ij}I.$$

Consequences: * If $i \neq j$, the matrices anti-commute $\rightarrow$ algebraic orthogonality.

- If $i = j$, we obtain $\sigma_i^2 = I$. This orthogonality ensures that Pauli matrices form a complete basis for Hermitian operators on $\mathbb{C}^2$.

Connection to quantum angular momentum

The angular momentum operators satisfy:

$$[L_x, L_y] = i\hbar L_z$$

and cyclic permutations. For a spin $-1/2$ particle:

$$S_x = \frac{\hbar}{2}\sigma_x, \quad S_y = \frac{\hbar}{2}\sigma_y, \quad S_z = \frac{\hbar}{2}\sigma_z.$$

Using the Pauli commutation relation: $[\sigma_x, \sigma_y] = 2i\sigma_z$, we obtain:

$$[S_x, S_y] = i\hbar S_z.$$

This shows that Pauli matrices realize the angular momentum algebra in the smallest possible representation. This is why they describe spin - $1/2$ systems and why qubit dynamics resemble rotations in three-dimensional space.

A concrete example: Rotating a spin- $1/2$ state around the x-axis and then the y-axis gives a different result than rotating around y first and then x. This non-commutativity is encoded directly in the Pauli commutators.

Eigenvalues and eigenstates

Each Pauli matrix has eigenvalues:

$$\lambda = +1, -1.$$

The corresponding eigenstates define the fundamental measurement directions:

- $\sigma_z : |0\rangle, |1\rangle$
- $\sigma_x : |+\rangle, |-\rangle$
- $\sigma_y : |+i\rangle, |-i\rangle$

These will become the poles and equatorial axes of the Bloch sphere.

Role of Pauli matrices in qubit dynamics

Any qubit Hamiltonian can be written as:

$$H = a_0 I + \vec{a} \cdot \vec{\sigma}.$$

The term $a_0 I$ contributes only a global phase. The vector $\vec{a}$ determines the actual dynamics. The time evolution operator is:

$$U(t) = e^{-iHt/\hbar} = e^{-i(\vec{a} \cdot \vec{\sigma})t/\hbar}.$$

This exponential describes a rotation of the qubit state around the axis $\vec{a}$ on the Bloch sphere.

This geometric interpretation will be developed in the next chapters.